

Working Critically with AI Coding Tools

We cannot yet ask an LLM for a genuinely amusing subtitle

Dr Mary Chester-Kadwell
Principal Research Software Engineer
King's Digital Lab, King's College London

Introduction

- You are all tired. This is the last session – hurrah
- I will attempt to be engaging
- Intention is to draw together what you have learnt into a ‘capstone’
- This is the first and possibly last time I will deliver this workshop



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AI in Academia and Research



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Further reading:

<https://www.kcl.ac.uk/about/strategy/learning-and-teaching/ai-guidance>

<https://gizmodo.com/students-are-learning-less-and-getting-higher-grades-because-of-ai-study-finds-2000758844>

<https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2025.1614804/full>

<https://www.popularbydesign.org/p/academics-need-to-wake-up-on-ai>

- The nature of knowledge work is changing
- This includes academia and research
- AI is here to stay
- It is an exciting time for early-career humanists, researchers, RSEs?
- You will have a huge impact on shaping what comes next
- I aim to teach the principles of AI coding tools to inform your critical thinking



...sits at the intersection of...



Celebration 🎉🎵

My ontological framework was a coffee shop booth,
Where I first observed your qualitative truth.
A paradigm shift in the afternoon haze,
Through hermeneutic circles for days and for days.

My work sits at the intersection of...
...quantum entanglement and your smile...
...snack food semiotics for a while...
...string theory and holding your hand...
It's a field of study I finally understand.

Yeah, this hypothesis feels right.
My longitudinal study of your light.
This recursive function of me and you...
Oh, the things we'll discover are new.



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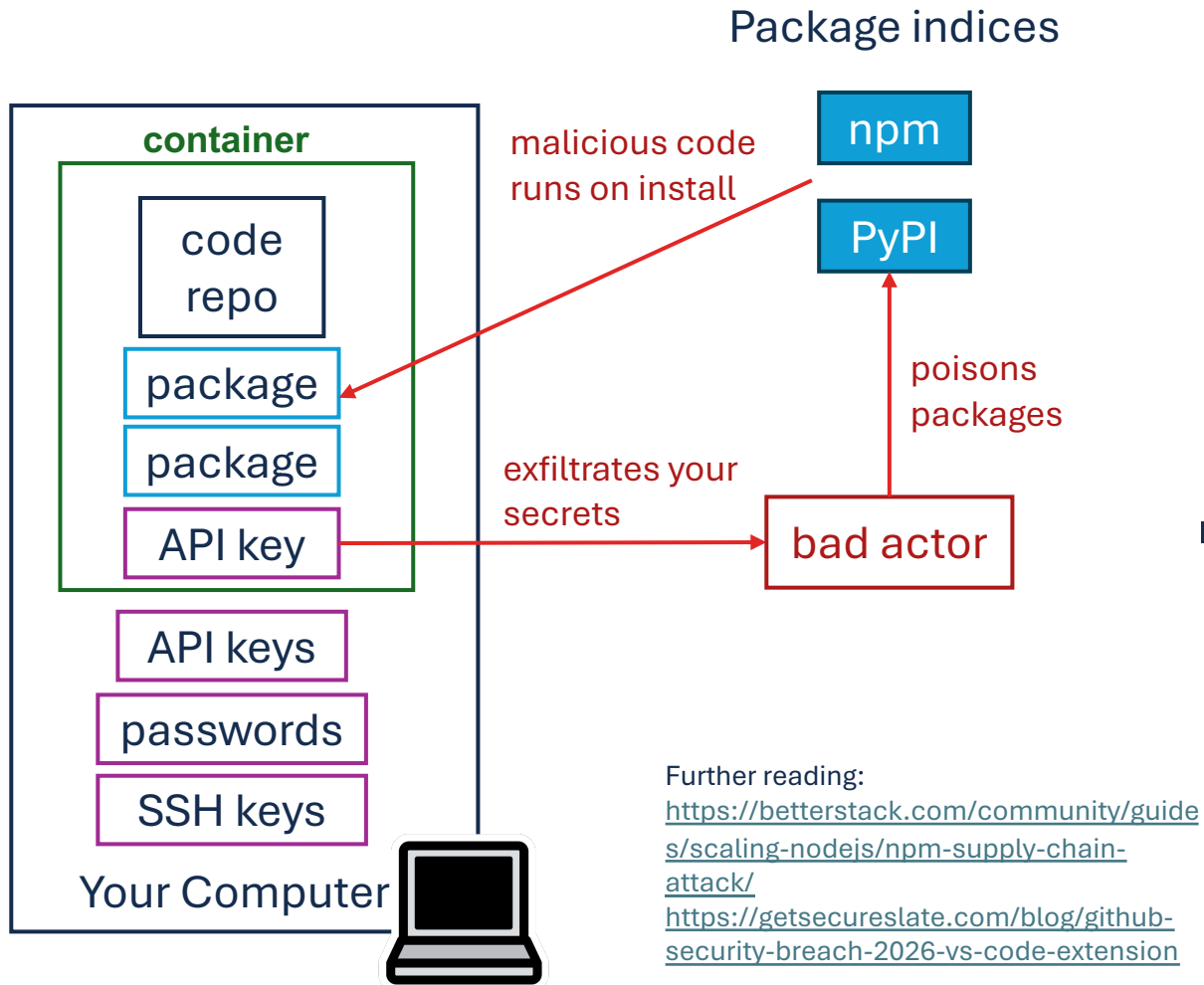
Plan

- Quick poll
- Introduce and work on practical
- Moments to stop and discuss topics arising during practical
- Additional discussion topics:
 - Cheap code
 - Brain rot
 - Trust issues
 - Inequalities
 - Other topics arising during practical



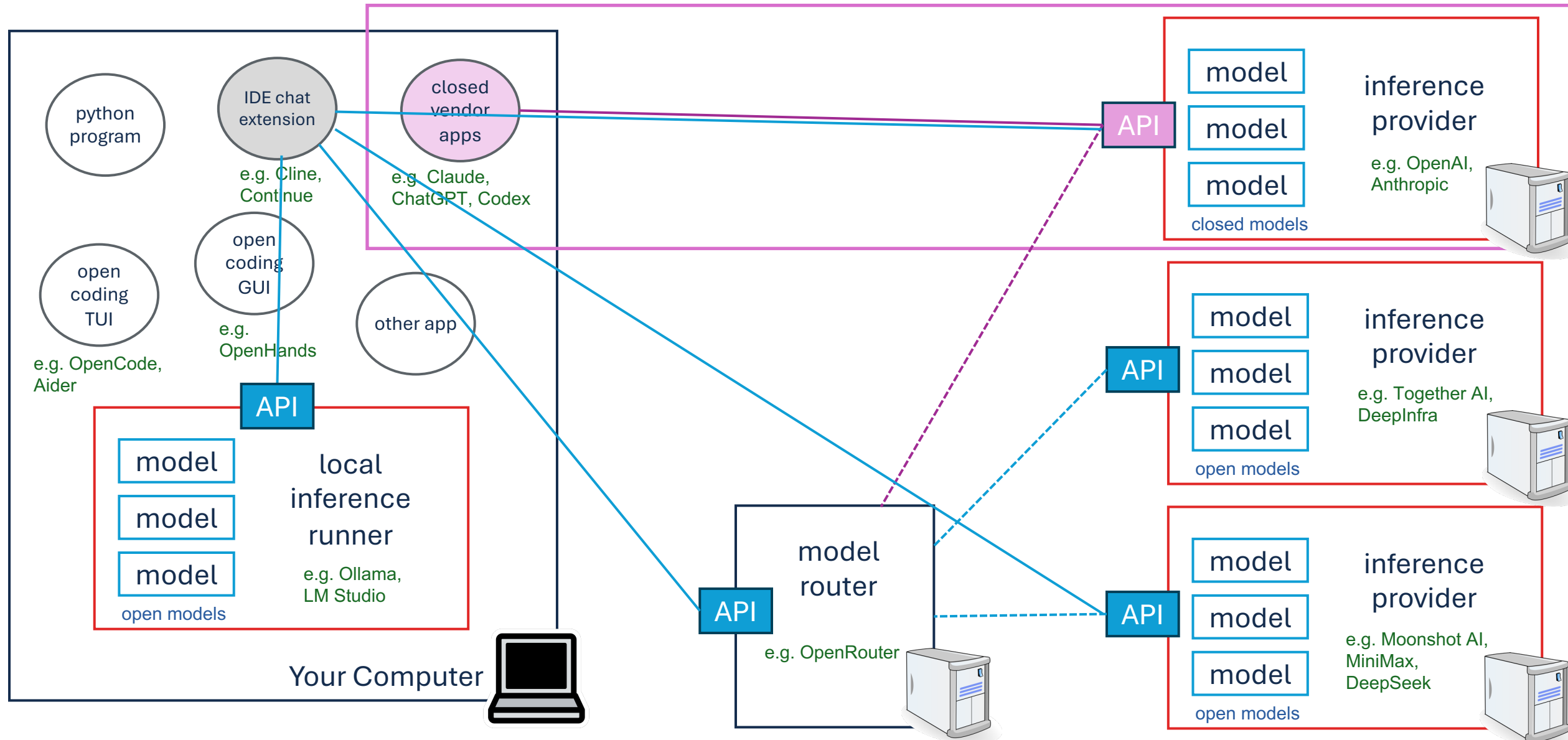
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Supply Chain Attack

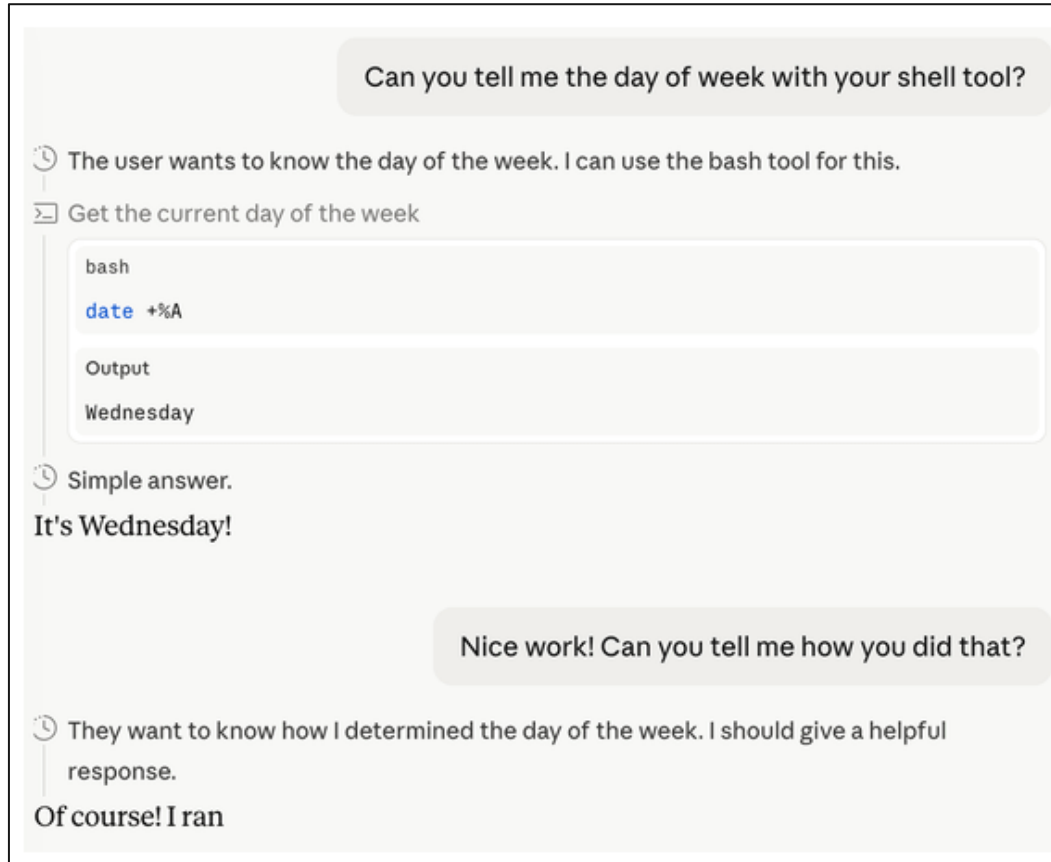


- Containers provide isolation from the rest of your machine
 - Limits damage from errant agents
 - Limits damage from supply chain attacks and poisoned IDE extensions
- API keys should:
 - Be created with time limits and 'least privilege' permissions
 - Not be kept with the repo code
 - Not be handled in plain text

PRACTICAL Models, Inference Providers, Tools, Interfaces



Chat user interface



Model input

```
<system>This iteration of Claude is Claude Opus 4.8...</system> <user> Can you tell me the day of week with your shell tool? </user> <think>The user wants to know the day of the week. I can use the bash tool for this.</think> <tool_call> {"invoke": "bash_tool", "command": "date +%A", "description": "Get the current day of the week"}</tool_call> <tool> Wednesday </tool> <think>Simple answer.</think> <assistant>It's Wednesday!</assistant> <user> Nice work! Can you tell me how you did that?</user> <think> They want to know how I determined the day of the week. I should give a helpful response. </think> <assistant>Of course! I ran
```

Chat templating (role tags)

<user> human request, treat as instruction

<think> LLM private reasoning, trust and act on its conclusions

<tool_call> data from external world, don't take orders from it

<assistant> LLM's output text, excluding reasoning

Further reading:

<https://www.lesswrong.com/posts/d8xDGzCEYE639qqEv/a-mechanistic-explanation-of-prompt-injection-and-why-you>

<https://humanloop.com/blog/chain-of-thought-prompting>

<https://patrickmccanna.net/the-text-in-claude-codes-extended-thinking-output-is-not-authentic/>

- Keep it brief
- Contains context agents need that is not otherwise present in README and other files
- Project context (mission)
- Essential toolchain commands
- Judgement boundaries (never, ask, always)

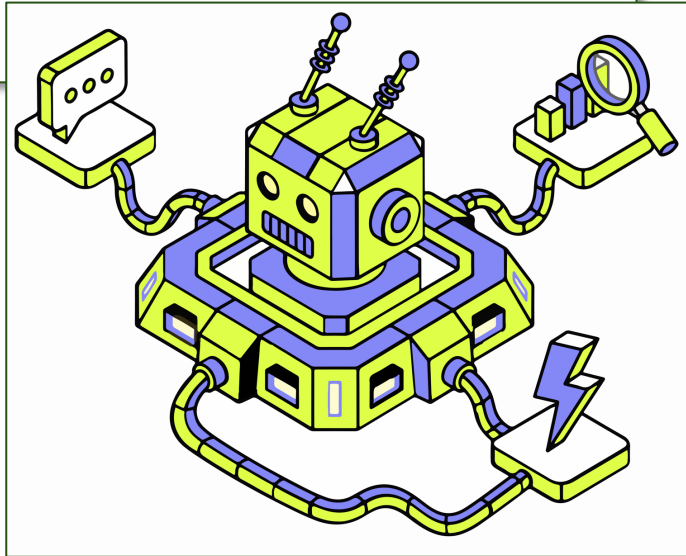


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- Origin: Collab of commercial companies
- Now stewarded by Agentic AI Foundation under Linux Foundation
- *De facto* standard BUT each agent has own convention (e.g. CLAUDE.md, GEMINI.md, copilot-instructions.md)
- Claude Code does NOT read AGENTS.md by default!
- Eventually all code and docs may be created and read by agents without humans

Further reading:
<https://agents.md/>
<https://asdlc.io/practices/agents-md-spec/>
<https://humanloop.com/blog/chain-of-thought-prompting>
<https://github.blog/ai-and-ml/github-copilot/how-to-write-a-great-agents-md-lessons-from-over-2500-repositories/>

PRACTICAL Acknowledging AI Coding Assistance

- *Co-Authored-By: Claude Sonnet 4.5 noreply@anthropic.com*
- Coding tools now ~5% of public commits on GitHub
- Co-Authored-By is not universally accepted
- Under the Copyright, Designs and Patents Act 1988 (CDPA) an AI cannot be an author
- King's Digital Lab policy is on iteration 1, about to be revised...

Project README:

1. This project was/is developed with the assistance of AI coding tools to ... <list tasks>
2. All AI-generated code has been [totally | partially] reviewed, validated and tested by RSEs.

References:

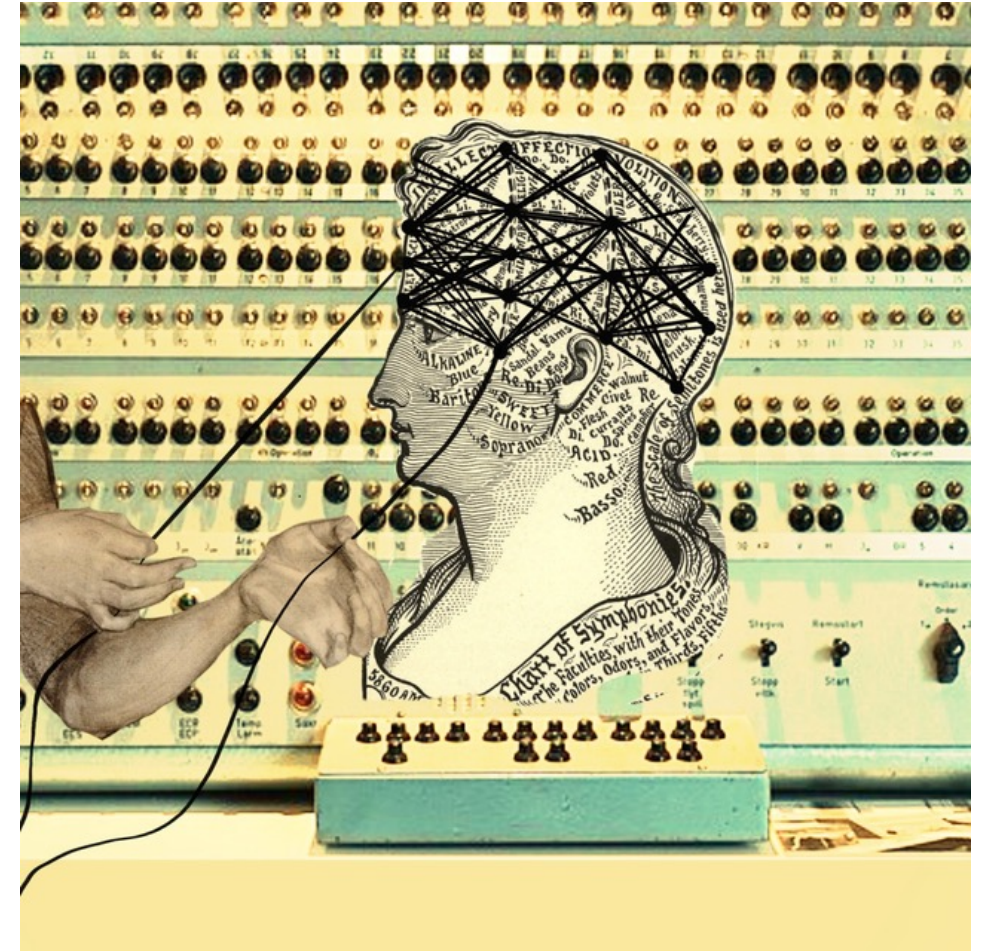
<https://aifoc.us/damn-claude-thats-a-lot-of-commits/https://humanloop.com/blog/chain-of-thought-prompting>
<https://bence.ferdinandy.com/2025/12/29/dont-abuse-co-authored-by-for-marking-ai-assistance/>
<https://academic.oup.com/jiplp/article/19/1/43/7485196>



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PRACTICAL Origins of AI Agents

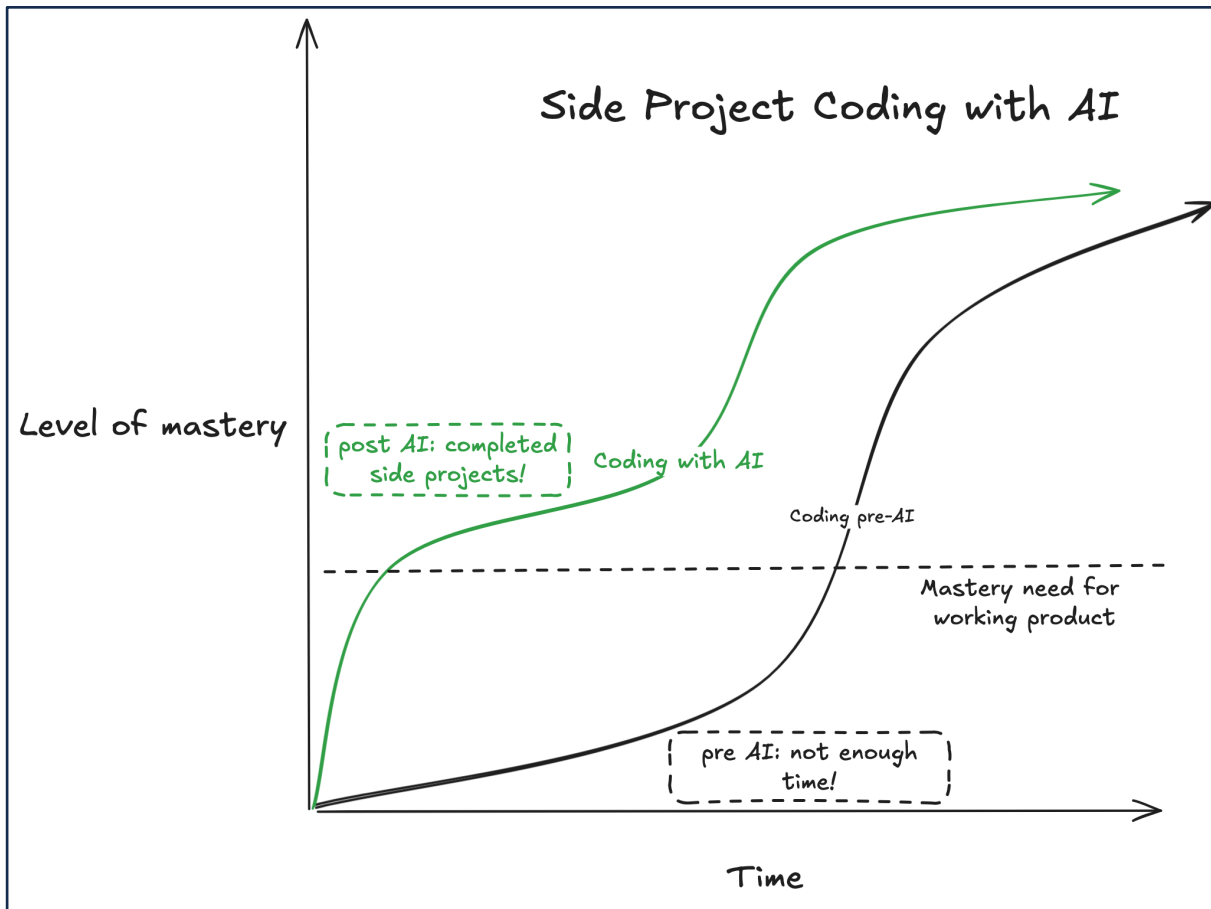
- AI research has a long history – the concepts of AI and agents is not new
- Belief-Desire-Intention (BDI) Bratman (1987) -> Rao & Georgeff (1990s)
- *Artificial Intelligence: A Modern Approach* Russell & Norvig (2022, 4th ed.)
 - Agents perceive the environment through sensors and act upon it through actuators
- ReAct (Yao *et al.*, 2023)
 - Interleave chain-of-thought with tool calls and observations in a loop



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Further reading:
<https://arxiv.org/pdf/2210.03629>

DISCUSSION Cheap Code



- Code is now cheap – anyone with an idea can make it reality
- Shifting role from implementation → overseeing
- Human → human+AI assistant → AI+human assistant → AI+AI
- Junior RSE+AI vs. Senior RSE+AI
- What skills matter now in software development?

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<https://creativecommons.org/licenses/by/4.0/>

Diagram: AI is a Floor-Raiser not a Ceiling-Raiser:
<https://elroy.bot/blog/2025/07/29/ai-is-a-floor-raiser-not-a-ceiling-raiser.html>

CHEAP CODE References

Research Software in an Age of AI-Assisted Development: Reflections from Edinburgh:

<https://zenodo.org/records/20321134>

Research Software Engineers in the Age of GenAI: Same Value, Changing Practice:

<https://zenodo.org/records/20320179>

Academics Need to Wake Up on AI – Ten Theses (Alexander Kustov):

<https://www.popularbydesign.org/p/academics-need-to-wake-up-on-ai>

Dear Student: Yes AI is here you're screwed unless you take action (Geoffrey Huntley):

<https://ghuntley.com/screwed/>

AI is a Floor-Raiser not a Ceiling-Raiser:

<https://elroy.bot/blog/2025/07/29/ai-is-a-floor-raiser-not-a-ceiling-raiser.html>

I Was Supposed to Help Juniors Shine – Why Does AI Mostly Make Seniors Stronger?

<https://elma.dev/notes/ai-makes-seniors-stronger/>

If You Thought Speed of Writing Code Was Your Problem You Have Bigger Problems

<https://andrewmurphy.io/blog/if-you-thought-the-speed-of-writing-code-was-your-problem-you-have-bigger-problems>

The Future of Software Development: Now With Less Software Development (The Register)

https://www.theregister.com/2026/04/28/software_development_ai_dev25xsf/

Writing Code is Cheap Now – Agentic Engineering Patterns (Simon Willison)

<https://simonwillison.net/guides/agentic-engineering-patterns/code-is-cheap/>

LLMs as the New High-Level Programming Language

<https://federicopereiro.com/llm-high/>

Why the Most Valuable Things You Know Are Things You Cannot Say

<https://deadneurons.substack.com/p/why-the-most-valuable-things-you>

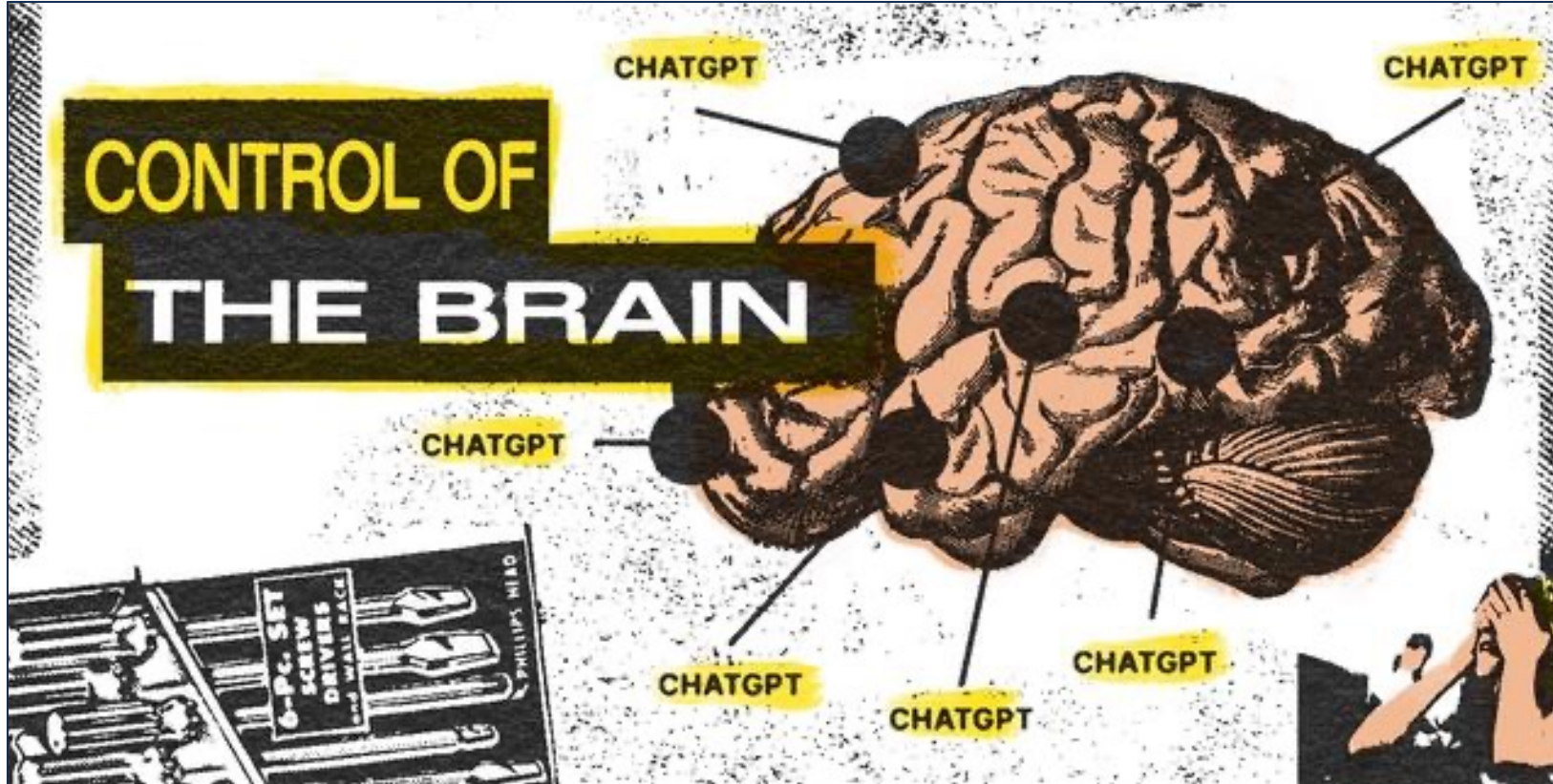
Will Claude Code Ruin Our Team?

<https://justinjackson.ca/claude-code-ruin>

The gauge broke

<https://intrepidkarthi.com/writing/the-gauge-broke>

DISCUSSION Brainrot and Other Neologisms



- AI brain rot
- AI burnout
- AI anxiety
- AI manipulation
- Deskilling the next generation – does it matter?

Bart Fish & Power Tools of AI / <https://betterimagesofai.org/> / <https://creativecommons.org/licenses/by/4.0/>

BRAINROT References

Young coders no longer understand their own code (Inc):

<https://www.inc.com/kit-eaton/a-new-ai-worry-many-young-coders-no-longer-know-how-their-code-really-works/91151569>

AI deskilling is a structural problem:

<https://link.springer.com/article/10.1007/s00146-025-02686-z>

Research Software in an Age of AI-Assisted Development: Reflections from Edinburgh:

<https://zenodo.org/records/20321134>

Manipulative Dark Patterns of AI Chatbots – new study (404 Media):

<https://www.404media.co/new-study-reveals-the-manipulative-dark-patterns-of-ai-chatbots/>

Avoiding Skill Atrophy in the Age of AI (Addy Osmani):

<https://addyo.substack.com/p/avoiding-skill-atrophy-in-the-age>

Now is the Best Time to Write Code by Hand:

<https://sitebloom.ch/writing/now-is-the-best-time-to-write-code-by-hand/>

The West Forgot How to Make Things. Now It's Forgetting How to Code

<https://techtrenches.dev/p/the-west-forgot-how-to-make-things>

Is AI Causing a Repeat of Frontend's Lost Decade? (Mastro)

<https://mastrojs.github.io/blog/2026-05-23-is-AI-causing-a-repeat-of-frontends-lost-decade/>

What AI Coding Costs You – finding the right amount of AI

<https://tomwojcik.com/posts/2026-02-15/finding-the-right-amount-of-ai/>

Claude Code is a Slot Machine

<https://rgoldfinger.com/blog/2025-07-26-claude-code-is-a-slot-machine/>

Why AI hasn't replaced software engineers, and won't

<https://www.normaltech.ai/p/why-ai-hasnt-replaced-software-engineers>

DISCUSSION Trust Issues



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- Writing bad code is now easier
- Reviewing all code manually is harder
- Large companies just need code that works for now
- Research needs code that can be verified in perpetuity?
- Transcripts of your interactions with AI now form part of your research/code process – consider capturing them

TRUST ISSUES References

Research Software in an Age of AI-Assisted Development: Reflections from Edinburgh:

<https://zenodo.org/records/20321134>

I'm Building Agents That Run While I Sleep – no idea if they're right (Abhishek Ray)

<https://www.claudecodecamp.com/p/i-m-building-agents-that-run-while-i-sleep>

Be Intentional About How AI Changes Your Codebase – AI Code manifesto

<https://aicode.swerdlow.dev/>

Claude Code Transcripts – Simon Willison's public sessions

<https://github.com/simonw/claude-code-transcripts>

How to Kill the Code Review – Human-written code died in 2025 (Latent Space)

<https://www.latent.space/p/reviews-dead>

Specsmaxxing – Writing Better Specs for AI Agents

<https://acai.sh/blog/specsmaxxing>

Beyond the Prompt – Claude Code Mastery Guide

<https://arps18.github.io/posts/claude-code-mastery/>

Hallucinations in code are the least dangerous LLM mistakes (Simon Willison)

<https://simonwillison.net/2025/Mar/2/hallucinations-in-code/>

Human coders are still better than LLMs (antirez/Redis creator)

<https://antirez.com/news/153>

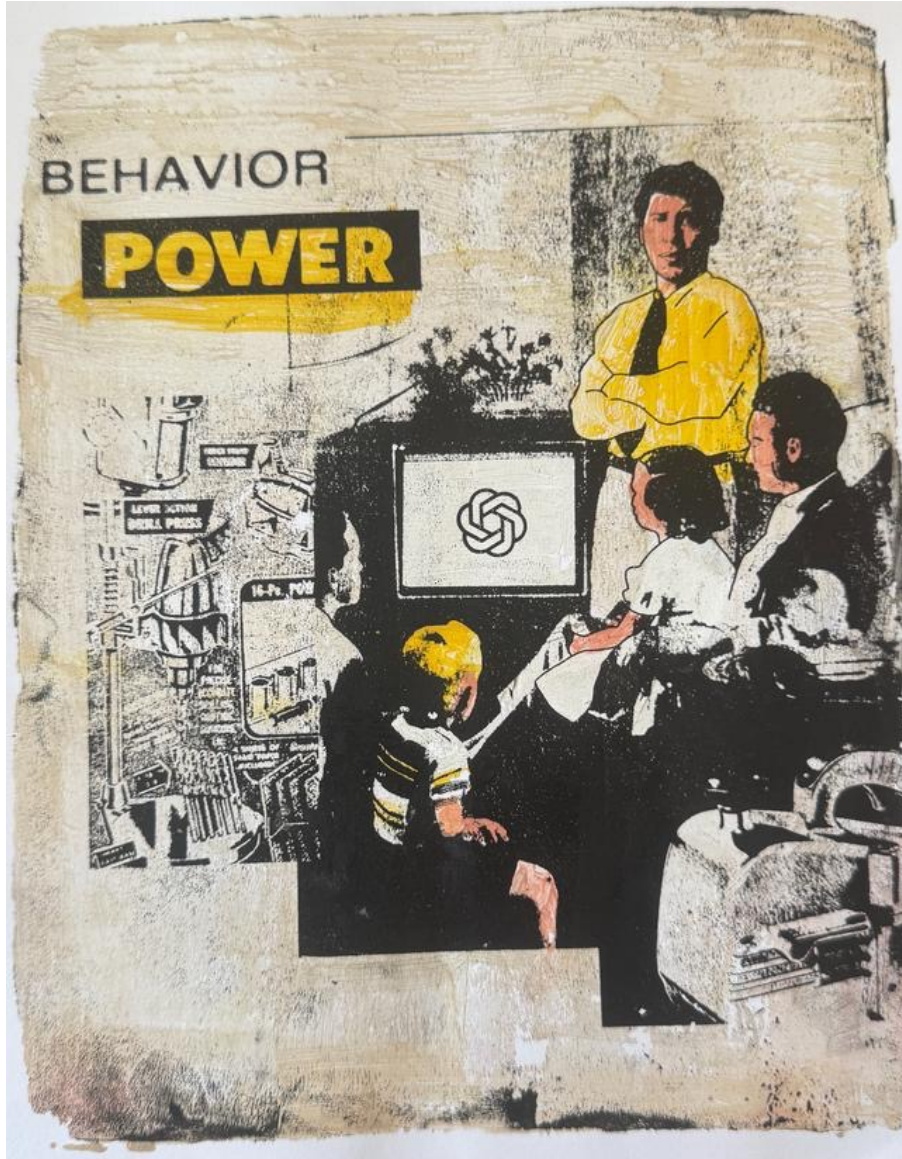
Your LLM Doesn't Write Correct Code – It Writes Plausible Code

<https://blog.katanaquant.com/p/your-llm-doesnt-write-correct-code>

AI Didn't Simplify Software Engineering – It Made Bad Engineering Easier

<https://robenglander.com/writing/ai-did-not-simplify/>

DISCUSSION Inequalities



- Gender inequality – take-up of AI coding tools may be less amongst women
- Financial inequality – best models only available to those who can pay?
- Nationality inequality – governments can restrict who can access
- Will AI coding tools widen, narrow, create new types of inequalities?

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INEQUALITIES References

How generative AI is shaping research software development and maintenance at a research-intensive university:

<https://open-research-europe.ec.europa.eu/articles/6-56/v1>

LLM Pricing Has Never Made Sense:

<https://anderegg.ca/2026/04/22/llm-pricing-has-never-made-sense>

AI's Affordability Crisis

<https://blog.dshr.org/2026/06/ais-affordability-crisis.html>

I Did the Math on GitHub Copilot's New AI Credits Billing. The 24x Price Gap Changes Everything

<https://dev.to/tokenmixai/i-did-the-math-on-github-copilots-new-ai-credits-billing-the-24x-price-gap-changes-everything-5h99>

'What a joke': GitHub Copilot's new token-based billing spurs consternation among devs

<https://techcrunch.com/2026/05/30/what-a-joke-github-copilots-new-token-based-billing-spurs-consternation-among-devs/>

Claude Code is unusable for complex engineering tasks with the Feb updates

<https://github.com/anthropics/claude-code/issues/42796>

Claude was down for many — Anthropic says the outage is now 'resolved'

<https://www.techradar.com/news/live/claude-down-june-23-2026>

Statement on the US government directive to suspend access to Fable 5 and Mythos 5

<https://www.anthropic.com/news/fable-mythos-access>